

Abdelrahman Abouroumia

AI & Backend Engineer

Email: aabouroumia@gmail.com | Phone: +20 110 047 1572

LinkedIn: [linkedin.com/in/abdelrahman-abouroumia](https://www.linkedin.com/in/abdelrahman-abouroumia) | GitHub: github.com/Ab-Romia | Hugging Face: huggingface.co/Ab-Romia

Portfolio: romia.dev | case studies, architecture breakdowns, and live demos

Experience

Software Engineer & Co-Founder

May 2025 to Present

Zaylon AI

- Owned the architecture, the code reviews and the deploys. Built the FastAPI backend behind them, with CI that load-tests the service nightly and smoke-probes the deployment every 30 minutes.
- Built the agent first as a multi-agent supervisor, then deliberately collapsed it into one LangGraph tool-calling agent with about 30 scoped tools, because one agent proved more reliable and far easier to evaluate.
- Made cross-tenant access structurally impossible rather than merely filtered, across about 80 tables, and added semgrep rules that fail the build when anyone writes a raw cross-tenant query.
- Wrote the dialect layer so a customer is answered in the language they actually typed, including Egyptian Arabic and Franco-Arabic written in Latin script, which general models handle badly.
- Verified every payment webhook by HMAC, and wrote a Postgres saga that unwinds a half-finished checkout in reverse.

AI Coding Expert

Sep 2024 to Feb 2026

Scale AI (Contract)

- Reviewed and ranked code across languages, for 18 months, in the RLHF pipelines that train frontier models.
- Wrote the test suites that decided whether a submission was correct, efficient, and honest about its edge cases.

Software Integration Engineering Intern

Jul 2025

Ejada

- Built core-banking REST APIs in Spring Boot, tuning the slow queries and adding Redis caching.
- Ran Kafka producers and consumers for async messaging; secured the gateway with OAuth2 and rate limiting on WSO2.

Automotive Cybersecurity Intern

Aug 2024 to Oct 2024

Swift-ACT

- Wrote threat-model catalogues and mitigations for CAN bus and ECU in-vehicle networks.
- Ran spoofing, replay and denial-of-service simulations in Dyna4 and CARLA, with ECU validation on Vector CANoe.

Projects

Talos: Team Chat Platform with a RAG Assistant

Graduation project, five-person team | [GitHub](#) | [Case Study](#)

- Owned the assistant: ingestion through a taskiq worker into Milvus, hybrid dense and BM25 retrieval fused with reciprocal rank fusion, a cross-encoder rerank, and answers that cite the passage they came from.
- The answers were weak, so I measured instead of guessing. I built an evaluation harness with a judge calibrated against my own blind grading, and proved a chunking fix moved judged correctness from 0.657 to 0.855.
- Containerized the worker and the scheduler for the team's Railway deploy, and ran Milvus on Zilliz Cloud.

commentdraft

Published package | [PyPI](#) | [GitHub](#) | [Case Study](#)

- Published a Python package on PyPI that triages comments on a creator's own posts into reply, skip or escalate, drafting each reply strictly from a source document so it declines what it cannot source.
- Made approval structural rather than configurable: no bulk-approve flag exists, one keystroke sends one reply, and a pytest proves keys typed before a draft renders approve nothing, after terminal typeahead defeated the first version.
- Wrote eight platform connection guides from primary sources, every claim carrying its URL and the date read, correcting published errors on the Instagram, TikTok, YouTube and Reddit comment APIs.

Virtual Banking Microservices

[GitHub](#) | [Case Study](#)

- Built a gateway and four Spring Boot services with a Kafka transfer saga: outbox, idempotent consumers, RS256 JWT.
- Proved the transfers were genuinely atomic with pessimistic locking, a balance constraint and Testcontainers tests.

ContextIQ: Hybrid-Retrieval RAG

[GitHub](#) | [Demo](#) | [Case Study](#)

- Built a retrieval service that runs entirely on CPU through ONNX, and made it abstain rather than guess when the retrieved context does not support an answer, tested against deliberately confusable distractors.

Multimodal Emotion Recognition

[GitHub](#) | [Demo](#) | [Case Study](#)

- Speaker-independent audio and facial pipeline on frozen WavLM-large, late fusion, unit-tested for leakage.

Education

Alexandria University, Faculty of Engineering

Graduated Jul 2026

B.Sc. in Computer & Communications Engineering (AI Concentration) | Cumulative GPA: 3.75 / 4.0

Technical Skills

Languages: Python, Java, SQL, JavaScript/TypeScript, C/C++

Backend: FastAPI, Spring Boot, Microservices, Event-Driven Architecture, Kafka, REST APIs, PostgreSQL, Redis

AI / ML: LangGraph, LangChain, PyTorch, RAG, LLM Agents, Agent Evaluation, LLM Orchestration, NLP, pgvector, Milvus

Infrastructure & Tooling: Docker, Railway, Git, GitHub Actions, CI/CD, Linux, pytest, semgrep, Testcontainers

Certifications: micro1 Certified Software Engineer | [Google AI Agents](#) | Meta Back-end | AWS Cloud & ML

Competitions: MTC-AIC3 BCI (Egypt National AI) | ECPC (3x) | Hepsiburada NLP Hackathon (TEKNOFEST)

Languages: English (Fluent) | Arabic (Native) | German (A2) | Spanish (Conversational)